

Algebraic Geometry



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Foreword

Chapter 1

Classical Algebraic Geometry

Chapter 2

Spectra

In this chapter the notion of spectra is introduced. This is a way to define a topology on the set of prime ideals of a ring. We discuss spectra of familiar rings and the connection of this definition to classical algebraic geometry. Further topological properties of spectra are discussed; the most important one is the fact that the spectrum of a ring is quasi-compact. Finally we prove that mapping a ring to its spectrum defines a functor from the category of rings to the category of topological spaces.

Recall our convention that rings are required to be commutative and unital, and every ring homomorphism is unital.

2.1 Spectrum of a Ring

Definition 2.1. Given a ring R , its **(prime) spectrum** $\operatorname{Spec} R$ is the set of its prime ideals.

It turns out that the spectrum $\operatorname{Spec} R$ comes with a natural topology. To define this topology, we introduce the following notation. Given a subset $S \subseteq R$, define

$$V(S) := \{\mathfrak{p} \in \operatorname{Spec} R : \mathfrak{p} \supseteq S\}.$$

It is straightforward to check that if $\mathfrak{a}$ is the ideal generated by S , then $V(S) = V(\mathfrak{a})$. Hence it suffices to consider only the sets $V(\mathfrak{a})$ whenever $\mathfrak{a}$ is an ideal of R . We shall also write $V(f_1, \dots, f_n)$ for $V(\langle f_1, \dots, f_n \rangle)$, where, as per usual, $\langle f_1, \dots, f_n \rangle$ denotes the ideal generated by $f_1, \dots, f_n$.

► **Exercise 2.2.** Show that for any ideal $\mathfrak{a}$ of R , $V(\sqrt{\mathfrak{a}}) = V(\mathfrak{a})$, where $\sqrt{\mathfrak{a}}$ denotes the radical of $\mathfrak{a}$.

The sets $V(\mathfrak{a})$ satisfy the following properties.

Proposition 2.3. *Let R be a ring, and let $\mathfrak{a}, \mathfrak{b}$ be ideals of R . Further, for each $k \in S$ let $\mathfrak{a}_k$ be ideals of R where S is some index set. Then*

1. *If $\mathfrak{a} \subseteq \mathfrak{b}$, then $V(\mathfrak{b}) \subseteq V(\mathfrak{a})$.*
2. *$V(\emptyset) = \text{Spec } R$ and $V(1) = \emptyset$.*
3. *$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.*
4. *$V(\sum_{k \in S} \mathfrak{a}_k) = \bigcap_{k \in S} V(\mathfrak{a}_k)$.*

► **Exercise 2.4.** Prove the above proposition. [Hint: the fact that $\sqrt{\mathfrak{a}\mathfrak{b}} = \sqrt{\mathfrak{a} \cap \mathfrak{b}}$ might be useful.]

The above proposition shows that the sets of the form $V(\mathfrak{a})$, where $\mathfrak{a}$ runs through all ideals of R , are closed sets of a topology on $\text{Spec } R$. This is the **Zariski topology** on $\text{Spec } R$. With the topology in mind, we will refer to elements of $\text{Spec } R$ as points. Given $x \in \text{Spec } R$, we shall write $\mathfrak{p}_x$ if we want to view x as a prime ideal of R .

There is a reverse correspondence of the map $V(-)$. If Y is a subset of $\text{Spec } R$, we define the ideal $I(Y)$ associated to Y by

$$I(Y) := \bigcap_{\mathfrak{p} \in Y} \mathfrak{p}.$$

Note that $I(\emptyset) = R$.

Proposition 2.5. *Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, and Y a subset of $\text{Spec } R$.*

1. *If $X \subseteq Y \subseteq \text{Spec } R$, then $I(Y) \subseteq I(X)$.*
2. *$\sqrt{I(Y)} = I(Y)$, i.e. $I(Y)$ is a radical ideal.*
3. *$I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ and $V(I(Y)) = \overline{Y}$ where $\overline{Y}$ is the topological closure of Y in $\text{Spec } R$.*

Proof. (1.) Trivial.

(2.) We only need to check the non-trivial inclusion $\sqrt{I(Y)} \subseteq I(Y)$. Suppose $a \in R$ and $n \geq 1$ satisfy $a^n \in I(Y)$. Then for each prime ideal $\mathfrak{p} \in Y$, $a^n \in \mathfrak{p}$. Then $a \in \mathfrak{p}$, and so $a \in I(Y)$.

(3.) The assertion that $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ follows from a theorem in commutative algebra which states that, for any ideal $\mathfrak{a}$ of R ,

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p} \text{ prime} \\ \mathfrak{p} \supseteq \mathfrak{a}}} \mathfrak{p}.$$

For the second assertion, note that $V(I(Y))$ contains Y . Now if $V(\mathfrak{b})$ is a closed set containing Y , then $I(Y) \supseteq I(V(\mathfrak{b})) = \sqrt{\mathfrak{b}} \supseteq \mathfrak{b}$ by the first assertion. It follows that

$$V(I(Y)) \subseteq V(\mathfrak{b}),$$

proving that $V(I(Y))$ is the closure of Y . □

The following corollaries are immediate.

Corollary 2.6. *The maps $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto V(\mathfrak{a})$ give rise to mutually inverse and inclusion-reversing bijections between the sets*

$$\{\text{radical ideals of } R\} \rightleftarrows \{\text{closed subsets of } \operatorname{Spec} R\}.$$

Corollary 2.7. *Let R be a ring and $\mathfrak{a}, \mathfrak{b}$ ideals of R . Then $V(\mathfrak{a}) = V(\mathfrak{b})$ if and only if $\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{b}}$.*

There is a connection between the sets $V(I)$ with the new definition in this chapter and the vanishing sets in the sense of classical algebraic geometry. For this, elements of R need to be interpreted as “functions” on $\operatorname{Spec} R$. Given $f \in R$, we define **the value of f at a prime $\mathfrak{p} \in \operatorname{Spec} R$** to be the image of f under the composition

$$R \rightarrow R/\mathfrak{p} \hookrightarrow \operatorname{Frac}(R/\mathfrak{p}).$$

Here we used the fact that $R/\mathfrak{p}$ is an integral domain so that its field of fractions may be constructed. Via this map, we see that $f(\mathfrak{p}) = 0$ if and only if $f \in \mathfrak{p}$. Hence $V(f)$ consists of prime ideals $\mathfrak{p}$ such that $f(\mathfrak{p}) = 0$. Therefore $V(f)$ is the *vanishing set* of f . Compare this situation to the classical case when R is the polynomial ring $k[x_1, \dots, x_n]$ and f is a polynomial in R .

Here we discuss spectra of familiar rings.

Example 2.8 (Spectrum of a Field). Every field k has only one prime ideal $\{0\}$. Thus $\operatorname{Spec} k$ consists of a single point.

Example 2.9 (Spectrum of $\mathbb{Z}$). The prime ideal of $\mathbb{Z}$ are the zero ideal (0) and the ideal generated by prime numbers (p) . Hence $\mathbb{Z}$ has infinitely many points.

More generally, the spectrum of an integral domain R is non-empty. It contains the point corresponding to the zero ideal. Let us denote this point by η . The fact that every prime ideal of R contains the zero ideal means that $\overline{\{\eta\}} = \operatorname{Spec} R$: the closure of the singleton $\{\eta\}$ is equal to the whole spectrum. Such a point is called a *generic point* of $\operatorname{Spec} R$ (Proposition 2.25). As strange as it is, generic points will turn out to be important to the theory of integral schemes.

For the last example in this section, we relate the coordinate ring $k[V]$ of an affine variety $V \subseteq \mathbb{A}^n$ to the spectrum $\operatorname{Spec} k[V]$ when k is an algebraically closed field.

Example 2.10. Let $V \subseteq \mathbb{A}^n$ be an affine variety given as the vanishing set $Z(f_1, \dots, f_i)$ of polynomials in $k[x_1, \dots, x_n]$. Let I denote the ideal consisting of polynomials vanishing on V . Recall that there is a one-to-one correspondence

$$\{\text{maximal ideals of } k[x_1, \dots, x_n]/I\} \iff \{\text{points of } V\}.$$

Let us denote the coordinate ring of V by $k[V] = k[x_1, \dots, x_n]/I$. If we write $\text{Spec}_M(A)$ for the set of maximal ideals of the ring A , then the points of V are in bijection with points of $\text{Spec}_M(k[V])$. The set $\text{Spec}_M(k[V])$ is a subspace of $\text{Spec}(k[V])$. It is easy to check that the subspace topology on $\text{Spec}_M(k[V])$ is the same one as the Zariski topology on V considered as a subspace of $k[x_1, \dots, x_n]$.

In topological terms maximal ideals are the closed points of the spectrum (see Proposition 2.25 below). Hence we “recover” classical algebraic geometry when we consider the closed points of the spectrum of a coordinate ring.

2.2 Topological Properties of Spectrums

Basis of $\text{Spec } R$

We now turn to the topological properties of $\text{Spec } R$ under the Zariski topology. Firstly, there is a basis for this topology which is derived from elements of R .

Definition 2.11. Let R be a ring and $f \in R$. We define

$$D(f) = \text{Spec } R \setminus V(f).$$

Plainly, $D(f)$ is open in $\text{Spec } R$. An open set of this form is called a **principal open subset** of $\text{Spec } R$.

The set $D(f)$ can also be regarded as the set of points where f does not vanish, that is, $D(f) = \{\mathfrak{p} \in \text{Spec } R : f(\mathfrak{p}) \neq 0\}$.

Proposition 2.12. *Let R be a ring. Principal open subsets of $\text{Spec } R$ form a basis of the Zariski topology.*

Proof. Let $U \subseteq \text{Spec } R$ be an open set, so that $U = \text{Spec } R \setminus V(\mathfrak{b})$ for some ideal $\mathfrak{b}$ of R . It is easy to see that

$$V(\mathfrak{b}) = \bigcap_{f \in \mathfrak{b}} V(f),$$

and consequently $U = \bigcup_{f \in \mathfrak{b}} D(f)$. □

The basis $\mathcal{B}$ consisting of principal open subsets is *stable under finite intersections*. This means that if $B_1, B_2 \in \mathcal{B}$, then $B_1 \cap B_2 \in \mathcal{B}$. This follows from the proposition below, and the fact that $D(0) = \emptyset$.

Proposition 2.13. *Let R be a ring. Then $D(f) \cap D(g) = D(fg)$ for all $f, g \in R$. Consequently the basis consisting of principal open subsets is stable under finite intersections.*

► **Exercise 2.14.** Prove the above proposition.

The following lemma is useful for turning a topological statement about a covering by principal open subsets into an algebraic statement.

Lemma 2.15 (“Covering Trick”). *Let $\{f_i\}_{i \in I}$ be a family of elements of a ring R and let $g \in R$. Then $D(g) \subseteq \bigcup_{i \in I} D(f_i)$ if and only if there is an integer $n > 0$ such that g^n is contained in the ideal $\mathfrak{a}$ generated by f_i .*

Proof. The condition $D(g) \subseteq \bigcup_{i \in I} D(f_i)$ is equivalent to $V(g) \supseteq \bigcap_{i \in I} V(f_i) = V(\mathfrak{a})$, which is equivalent to $g \in \sqrt{\mathfrak{a}}$. $\square$

Two special cases of the above theorem are often used. Firstly, $D(g) \subseteq D(f)$ if and only if $g \in \sqrt{(f)}$, if and only if $g^n = fa$ for some $a \in R$. This is important when we define the structure sheaf of $\text{Spec } R$.

Secondly, setting $g = 1$ shows that the collection $\{D(f_i)\}_{i \in I}$ covers $\text{Spec } R$ if and only if $1 \in \langle f_i : i \in I \rangle$. Hence there is a finite subset $J \subseteq I$ such that $1 = \sum_{j \in J} r_j f_j$ where $r_j \in R$. This implies that $\text{Spec } R = \bigcup_{j \in J} D(f_j)$. Thus we have found a finite subcover of $\{D(f_i)\}_{i \in I}$, implying that $\text{Spec } R$ is quasi-compact. More generally, every principal open subset of $\text{Spec } R$ is quasi-compact by the same argument.

Proposition 2.16. *Let R be a ring. Then every principal open subset $D(g)$, $g \in R$, is quasi-compact.*

► **Exercise 2.17.** Prove this proposition by adapting the proof that $\text{Spec } R$ is quasi-compact.

Irreducibility

A non-empty topological space is **irreducible** if it cannot be written as a union of two proper closed subsets. In other words, a topological space $X \neq \emptyset$ is irreducible if and only if whenever $X = C_1 \cup C_2$, where C_1 and C_2 are closed subsets, then either $C_1 = X$ or $C_2 = X$. A topological space is **reducible** if it is not irreducible. It is useful to consider the empty topological space as being reducible.

A subset of a topological space is irreducible if it is irreducible under the subspace topology. Similar to connected components, we define an **irreducible component** to be a subspace $Z \subseteq X$ such that Z is maximally irreducible subset of X , that is, Z is irreducible and whenever Y is irreducible and $Z \subseteq Y$, then $Z = Y$.

We collect some properties of irreducible spaces here, whose proofs are mechanical.

Proposition 2.18. *The following are equivalent for a non-empty topological space X .*

1. X is irreducible.
2. Every non-empty open subset of X is dense in X .
3. Every two non-empty open subsets of X has non-empty intersection.
4. Every non-empty open subset of X is irreducible.

► **Exercise 2.19.** Prove the above proposition.

Proposition 2.20. *Let $f: X \rightarrow Y$ be a continuous map between topological spaces. If $Z \subseteq X$ is an irreducible subset, then its image $f(Z)$ is irreducible.*

Proof. Clearly $f(Z)$ is non-empty because Z is. Suppose $f(Z) = C_1 \cup C_2$ where C_1 and C_2 are closed subsets of $f(Z)$ under the subspace topology. Hence $C_1 = D_1 \cap f(Z)$ and $C_2 = D_2 \cap f(Z)$ for some closed sets D_1 and D_2 of Y . We then have

$$Z = (f^{-1}(D_1) \cap Z) \cup (f^{-1}(D_2) \cap Z).$$

This is a decomposition of Z into two closed subsets. Since Z is irreducible, either $f^{-1}(D_1) \cap Z = Z$ or $f^{-1}(D_2) \cap Z = Z$. Hence either $C_1 = f(Z)$ or $C_2 = f(Z)$. $\square$

Proposition 2.21. *Let X be a topological space. Then $Y \subseteq X$ is irreducible if and only if its closure $\overline{Y}$ is irreducible.*

Proof. First assume that $Y \subseteq X$ is irreducible. Then $\overline{Y}$ is clearly non-empty. If $\overline{Y} = C_1 \cup C_2$ for some closed subsets C_1, C_2 of X , then $Y = (C_1 \cap Y) \cup (C_2 \cap Y)$. Since Y is irreducible, one of the subsets must be equal to Y , say $C_1 \cap Y = Y$. Hence C_1 is a closed set containing Y , and so $\overline{Y} \subseteq C_1$. Plainly, $C_1 \subseteq \overline{Y}$ and therefore $\overline{Y} = C_1$.

For the converse, assume that Y is reducible. If Y is empty, then $\overline{Y} = \emptyset$ is also reducible. Hence we may assume that Y is non-empty and write $Y = C_1 \cup C_2$, where C_1 and C_2 are proper closed subsets of Y under the subspace topology. Then $\overline{Y} = \overline{C_1} \cup \overline{C_2}$ by the property of closure. If $\overline{Y} = \overline{C_1}$, then

$$C_1 = \overline{C_1} \cap Y = \overline{Y} \cap Y = Y,$$

a contradiction. Similarly we conclude that $\overline{Y} \neq \overline{C_2}$. Therefore we have found a decomposition of $\overline{Y}$ into two proper closed subsets. $\square$

We now turn to irreducible subsets of $\text{Spec } R$, when R is a ring. The starting point is the following proposition which says that irreducible subsets are related to prime ideals.

Proposition 2.22. *A subset $Y \subseteq \text{Spec } R$ is irreducible if and only if $I(Y)$ is a prime ideal of R .*

Proof. Assume that Y is irreducible. Then $\overline{Y}$ is also irreducible. Let $fg \in I(Y) \subseteq R$. Using Proposition 2.3 and Proposition 2.5, we have the identities

$$V(f) \cup V(g) = V(fg) \supseteq V(I(Y)) = \overline{Y}.$$

Hence $\overline{Y} = (V(f) \cap \overline{Y}) \cup (V(g) \cap \overline{Y})$. Since $\overline{Y}$ is irreducible, it must be equal to one of the subsets, say $\overline{Y} = V(f) \cap \overline{Y}$. Therefore $\overline{Y} \subseteq V(f)$. Again, by Proposition 2.5, we have $I(Y) \supseteq \sqrt{(f)}$. This implies that $f \in I(Y)$. This proves that $I(Y)$ is prime.

Now assume that $\mathfrak{p} = I(Y)$ is prime. By Proposition 2.5, $\overline{Y} = V(\mathfrak{p})$. Let $V(\mathfrak{a}), V(\mathfrak{b})$ be closed sets such that

$$V(\mathfrak{p}) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}).$$

Recall that $\mathfrak{p} = \sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$ by commutative algebra. Since $\mathfrak{p}$ is prime, either $\mathfrak{p} = \sqrt{\mathfrak{a}}$ or $\mathfrak{p} = \sqrt{\mathfrak{b}}$. Hence $V(\mathfrak{p}) = V(\mathfrak{a})$ or $V(\mathfrak{p}) = V(\mathfrak{b})$. Thus $V(\mathfrak{p}) = \overline{Y}$ is irreducible, and so does Y . $\square$

Proposition 2.23. *There is a one-to-one correspondence between $\text{Spec } R$ and the set of closed irreducible subsets of $\text{Spec } R$ given in one direction by $\mathfrak{p} \mapsto \overline{\{\mathfrak{p}\}}$. Under this correspondence, minimal prime ideals of R correspond bijectively to irreducible components of $\text{Spec } R$.*

Proof. If $\mathfrak{p}$ is prime, then $I(\{\mathfrak{p}\}) = \mathfrak{p}$ and thus $\overline{\{\mathfrak{p}\}} = V(I(\{\mathfrak{p}\})) = V(\mathfrak{p})$ is irreducible by Proposition 2.22. Now if Y is a closed irreducible subset of $\text{Spec } R$, then by Proposition 2.22, $I(Y)$ is a prime ideal. Hence the correspondence is given by the functions $V(-)$ and $I(-)$ which are proved to be mutually inverse bijections in Corollary 2.6.

The second statement follows immediately from the observation that $\overline{\{\mathfrak{p}\}} \subseteq \overline{\{\mathfrak{q}\}}$ if and only if $\mathfrak{p} \supseteq \mathfrak{q}$ for all prime ideals $\mathfrak{p}, \mathfrak{q}$ of R . $\square$

Special Points

Definition 2.24. Let X be a topological space.

1. A point $x \in X$ is a **closed point** if $\{x\}$ is closed.
2. A point $\eta \in X$ is a **generic point** if $\overline{\{\eta\}} = X$.
3. Let x and x' be points of X . We say that x is a **generization** of x' and that x' is a **specialisation** of x if $x' \in \overline{\{x\}}$.
4. A point $x \in X$ is a **maximal point** of X if $\overline{\{x\}}$ is an irreducible component of X .

For spectra the above topological notions corresponds to the following algebraic statements.

Proposition 2.25. *Let R be a ring.*

1. *A point $x \in \operatorname{Spec} R$ is closed if and only if the ideal $\mathfrak{p}_x$ is maximal.*
2. *A point $\eta \in \operatorname{Spec} R$ is generic if and only if the ideal $\mathfrak{p}_\eta$ is the unique minimal prime ideal.*
3. *A point x is a generization of x' if and only if $\mathfrak{p}_x \subseteq \mathfrak{p}_{x'}$.*
4. *A point $x \in \operatorname{Spec} R$ is maximal if and only if $\mathfrak{p}_x$ is a minimal prime ideal.*

► **Exercise 2.26.** Prove the proposition.

2.3 Spec as a Functor

We have seen that a ring R gives rise to the topological space $\operatorname{Spec} R$. In addition to this, a ring homomorphism $\varphi: R \rightarrow S$ gives rise to a continuous map between the underlying topological space $\operatorname{Spec} S \rightarrow \operatorname{Spec} R$, as we shall see in the following lemma.

Lemma 2.27. *Let $\varphi: R \rightarrow S$ be a ring homomorphism. Then there is a continuous map $\operatorname{Spec} \varphi: \operatorname{Spec} S \rightarrow \operatorname{Spec} R$ given by $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$.*

In particular, the identity $(\operatorname{Spec} \varphi)^{-1}(V(\mathfrak{b})) = V(\varphi(\mathfrak{b}))$ holds for all ideals $\mathfrak{b}$ of R .

Proof. It is an easy exercise in commutative algebra to show that $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of R if $\mathfrak{q}$ is a prime ideal of S . To prove that $f := \operatorname{Spec} \varphi$ is continuous, let $V(\mathfrak{b})$ be a closed subset of $\operatorname{Spec} R$. Then

$$f^{-1}(V(\mathfrak{b})) = \{\mathfrak{p} \in \operatorname{Spec} S : \mathfrak{b} \subseteq \varphi^{-1}(\mathfrak{p})\},$$

from the definition, as one can check. But $\mathfrak{b} \subseteq \varphi^{-1}(\mathfrak{p})$ if and only if $\varphi(\mathfrak{b}) \subseteq \mathfrak{p}$, and thus $f^{-1}(V(\mathfrak{b})) = V(\varphi(\mathfrak{b}))$ as claimed. Since every preimage under f of a closed set is closed, this shows that f is continuous. $\square$

We left the reader to check that, for any ring R and the identity map $\operatorname{id}_R: R \rightarrow R$, the map $\operatorname{Spec}(\operatorname{id}_R)$ is the identity function of $\operatorname{Spec} R$. Moreover, we have $\operatorname{Spec}(\psi \circ \varphi) = \operatorname{Spec} \varphi \circ \operatorname{Spec} \psi$ for any ring homomorphisms $\varphi: R \rightarrow S$ and $\psi: S \rightarrow T$. Hence Spec is a contravariant functor from the category of rings to the category of topological spaces.

A continuous map between topological spaces $f: X \rightarrow Y$ is **dominant** if $f(X)$ is dense in Y . For the morphism $\operatorname{Spec} \varphi: \operatorname{Spec} S \rightarrow \operatorname{Spec} R$ between spectra, this happens precisely when $\ker \varphi$ is contained in the nilradical of R .

Corollary 2.28. *Let $\varphi: R \rightarrow S$ be a ring homomorphism. Then $\operatorname{Spec} \varphi$ is dominant if and only if $\ker \varphi \subseteq \operatorname{Nil}(R)$.*

Proof. See Problem 2.8. $\square$

Proposition 2.29. *Let $\varphi: R \rightarrow S$ be a surjective homomorphism of rings. Then $\text{Spec } \varphi$ defines a homeomorphism from $\text{Spec } S$ to the set $V(\mathfrak{a})$ of $\text{Spec } R$.*

Proof. content... □

Since every quotient homomorphism is surjective, we obtain the following corollary.

Corollary 2.30 (Homeomorphism between $\text{Spec } R/\mathfrak{a}$ and $V(\mathfrak{a})$). *Let R be a ring, $\mathfrak{a}$ its ideal, and $\pi: R \rightarrow R/\mathfrak{a}$ the quotient homomorphism. Then $\text{Spec } \pi$ is a homeomorphism from $\text{Spec } R/\mathfrak{a}$ onto the set $V(\mathfrak{a})$.*

Proposition 2.31. *Let R be a ring, $S \subseteq R$ a multiplicative subset, and $\psi: R \rightarrow S^{-1}R$ the canonical homomorphism $r \mapsto r/1$. Then $\text{Spec } \psi$ is a homeomorphism from $\text{Spec}(S^{-1}R)$ onto the set of prime ideals $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$.*

2.4 Problems

2.1 Prove that, for any ring R and any ideal I of R , $V(I) = \emptyset$ if and only if $I = R$.

2.2 Prove the following theorem from commutative algebra: if R is a ring and $\mathfrak{a}$ is an ideal of R , then

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p} \text{ prime} \\ \mathfrak{p} \supseteq \mathfrak{a}}} \mathfrak{p}.$$

2.3 Show that if $f \in R$ vanishes at every point of $\text{Spec } R$ (that is, $f(\mathfrak{p}) = 0$ for all $\mathfrak{p} \in \text{Spec } R$), then f is nilpotent.

2.4 Prove that $\text{Spec } R$ is irreducible if and only if the nilradical of R is prime.

2.5 Show that if $\mathfrak{p} \subseteq R$ is a prime ideal, then $\{\mathfrak{p}\}$ is the only generic point of $V(\mathfrak{p})$.

2.6 Show that if R is an integral domain, then $\text{Spec } R$ is irreducible.

2.7 Given a ring homomorphism $\varphi: R \rightarrow S$, let f denote the corresponding continuous map $\text{Spec } \varphi: \text{Spec } S \rightarrow \text{Spec } R$. Show that

- (a) $f^{-1}(D(r)) = D(\varphi(r))$ for all $r \in R$.
- (b) $V(\varphi^{-1}(\mathfrak{b})) = \overline{f(V(\mathfrak{b}))}$ for all ideal $\mathfrak{b} \subseteq S$.

2.8 Prove Corollary 2.28. [Hint: Use Problem 2.7].

2.9 Let R be a ring and $\mathfrak{p}$ a prime ideal of R .

- (a) Prove that $\text{Spec } R_{\mathfrak{p}}$ has a unique closed point.
- (b) Prove that if U is an open subset which contains the unique closed point of $\text{Spec } R_{\mathfrak{p}}$, then $U = \text{Spec } R_{\mathfrak{p}}$.

Chapter 3

Sheaves and Affine Schemes

3.1 Presheaves

Definition 3.1. Let X be a topological space. A **presheaf of Abelian groups** $\mathcal{F}$ on X consists of the following data:

1. an assignment of an Abelian group $\mathcal{F}(U)$ to each open subset $U \subseteq X$;
2. for each inclusion of open subsets $U \subseteq V$, there is a group homomorphism $\text{res}_U^V: \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ such that
 - (a) $\text{res}_U^U = \text{id}_{\mathcal{F}(U)}$ for all open subset U ;
 - (b) if $U \subseteq V \subseteq W$ are open subsets, then $\text{res}_U^W = \text{res}_U^V \circ \text{res}_V^W$.

An element of $\mathcal{F}(U)$ is called a **section** of $\mathcal{F}$ over U . The morphism res_U^V is called the **restriction map** from V to U . If $s \in \mathcal{F}(V)$, the more suggestive notation for $\text{res}_U^V(s)$ is $s|_U$. The condition for restriction maps is then $(s|_V)|_U = s|_U$ for all open subsets $U \subseteq V \subseteq W$ and $s \in \mathcal{F}(W)$.

Example 3.2. An example of presheaves on a topological space is the presheaf of real-valued smooth function on a smooth manifold M . If $U \subseteq M$ is open, we define $\mathcal{C}^\infty(U)$ to be the set of real-valued smooth functions on U . The restriction maps are simply restriction of functions, which is the motivation for the notation $s|_U$.

We now turn to the definition of morphisms between presheaves.

Definition 3.3. Given two presheaves $\mathcal{F}$ and $\mathcal{G}$, a **morphism of presheaves** $f: \mathcal{F} \rightarrow \mathcal{G}$ is the collection of maps $f_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for each open subset $U \subseteq X$ such that whenever $U \subseteq V$ are open subsets, the diagram

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{f_V} & \mathcal{G}(V) \\ \text{res}_U^V \downarrow & & \downarrow \text{res}_U^V \\ \mathcal{F}(U) & \xrightarrow{f_U} & \mathcal{G}(U) \end{array}$$

commutes.

We can alternatively give a categorical perspective on presheaves. Let X be a topological space. The open sets of X form a category $(\text{Ouv}(X))$ of open subsets of X . We declare that there is exactly one morphism $U \rightarrow V$ in $(\text{Ouv}(X))$ if and only if $U \subseteq V$. Put simply, this is the order category of open subsets of X with the ordering given by inclusion of sets. Then a presheaf on X is simply a contravariant functor $\mathcal{F}: (\text{Ouv}(X))^{\text{op}} \rightarrow (\text{Ab})$. A morphism of presheaves $f: \mathcal{F} \rightarrow \mathcal{G}$ is then a natural transformation between the two functors.

By changing the target so that $\mathcal{F}: (\text{Ouv}(X))^{\text{op}} \rightarrow \mathcal{C}$ maps to another category $\mathcal{C}$, we obtain a presheaf $\mathcal{F}$ on X with value in $\mathcal{C}$, not just Abelian groups. In the following we mostly consider the following types of presheaves:

1. Presheaves of sets $(\text{Ouv}(X))^{\text{op}} \rightarrow (\text{Set})$.
2. Presheaves of Abelian groups $(\text{Ouv}(X))^{\text{op}} \rightarrow (\text{Ab})$.
3. Presheaves of rings $(\text{Ouv}(X))^{\text{op}} \rightarrow (\text{Ring})$.
4. Given a ring R , presheaves of R -modules $(\text{Ouv}(X))^{\text{op}} \rightarrow (R\text{-Mod})$ and presheaves of R -algebras $(\text{Ouv}(X))^{\text{op}} \rightarrow (R\text{-Alg})$.

Example 3.4 (Constant Presheaf). Let A be a fixed Abelian group and let X be a topological space. We can define a presheaf $\mathcal{A}$ on X by setting $\mathcal{A}(U) = A$ for all open subsets $U \subseteq X$ and all restriction maps are the identity homomorphism. This is called the **constant presheaf** with value A on X .

Example 3.5 (Restriction of Presheaf). Let X be a topological space and $\mathcal{F}$ a presheaf on X . If U is an open subset of X , then the assignment

$$\mathcal{F}|_U(V) = \mathcal{F}(V) \quad \text{for each open subset } V \subseteq U$$

define a presheaf on U . The presheaf $\mathcal{F}|_U$ is called the restriction of the presheaf $\mathcal{F}$ onto U .

Example 3.6 (Sub-presheaf). Let $\mathcal{F}$ and $\mathcal{G}$ be a presheaf on a topological space X with values in algebraic objects such as Abelian groups or rings. If $\mathcal{F}(U)$ is a subgroup of $\mathcal{G}(U)$ for all open $U \subseteq X$, then $\mathcal{F}$ is said to be a **sub-presheaf** of $\mathcal{G}$.

3.2 Sheaves

The fundamental property of functions on a topological space X is that we can glue functions $f_i: U_i \rightarrow X$ defined on open subsets $U_i \subseteq X$ together into a function $f: U \rightarrow X$ defined on the union $U = \bigcup_i U_i$ of the open sets. Since we like to think of sections on U as “functions” on U , the above property should hold for the type of presheaves which are desirable for our study. Hence we are led to the following definition.

Definition 3.7. Let $\mathcal{F}$ be a presheaf of Abelian groups on a topological space X . We say that $\mathcal{F}$ is a **sheaf** if it satisfies the following two properties: for all open subsets $U \subseteq X$ and for all open coverings $\{U_i\}_{i \in I}$ of U ,

1. (Uniqueness) Let $s, t \in \mathcal{F}(U)$ be sections such that $s|_{U_i} = t|_{U_i}$ for all $i \in I$. Then $s = t$.
2. (Gluing) Given a collection of sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, there exists $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all $i \in I$.

The above two properties are called the **sheaf axioms**. Given a collection of sections $s_i \in \mathcal{F}(U_i)$ and $s \in \mathcal{F}(U)$ as in the second part, the section s is said to be obtained by **gluing the sections** s_i . It is unique by the first part.

Note that we may replace the target category from the category of Abelian groups to other categories, as noted in the previous section. For sheaves with values in a category of algebraic structures, the sheaf axioms can be restated in terms of exact sequences as follows.

Lemma 3.8. Let $\mathcal{F}$ be a sheaf of Abelian groups (or rings, R -modules, R -algebras with R a ring) on a topological space X . $\mathcal{F}$ is a sheaf if and only if for every open subset $U \subseteq X$ and open cover $\{U_i\}_{i \in I}$ of U , the sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\varphi} \prod_{i \in I} \mathcal{F}(U_i) \xrightarrow{\sigma} \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j),$$

is exact, where $\varphi: s \mapsto (s|_{U_i})_{i \in I}$ and $\sigma: (s_i)_{i \in I} \mapsto (s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j})_{i,j}$.

Proof. The uniqueness condition is equivalent to the sequence being exact at $\mathcal{F}(U)$. Similarly, the gluing condition is equivalent to the sequence being exact at $\prod_i \mathcal{F}(U_i)$. $\square$

A **morphism of sheaves** is a morphism of presheaves. We then obtain the category of sheaves on a topological space X , denoted by $(\text{Sh}(X))$. Note that we really should have

chosen a notation such as $(\mathbf{AbSh}(X))$ to unambiguously clarify that it is the category of sheaves of *Abelian groups* on X that we are considering. Still, in this section and wherever appropriate, the proof works for any target category of sheaves. With this in mind, we shall prove most statement for sheaves of Abelian groups and left the other cases for the reader.

The category of presheaves on X is denoted by $(\mathbf{PSh}(X))$. Clearly $(\mathbf{Sh}(X))$ is a full subcategory of $(\mathbf{PSh}(X))$.

Remark 3.9. For a sheaf $\mathcal{F}$ on a topological space X , $\mathcal{F}(\emptyset)$ is precisely the final object in the respective category. This can be seen from Lemma 3.8 by considering the empty covering of the empty set. Since the empty product is the terminal object, we obtain the exact sequence $0 \rightarrow \mathcal{F}(\emptyset) \rightarrow 0 \rightarrow 0$, showing that $\mathcal{F}(\emptyset) \cong 0$.

When $\mathcal{F}$ is a sheaf of sets, we can argue directly from the definition of sheaves that $\mathcal{F}(\emptyset)$ is a set with one element. Note that Lemma 3.8 does not apply to this case.

Presheaves and sheaves can be restricted onto any open subset.

Definition 3.10. Suppose we have a presheaf $\mathcal{F}$ on a topological space X . If $U \subseteq X$ is an open subset, then the presheaf $\mathcal{F}|_U$ defined by $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for all open subset $V \subseteq U$ is a presheaf on U called the **restriction** of $\mathcal{F}$ to U .

Clearly $\mathcal{F}|_U$ is a sheaf if $\mathcal{F}$ is a sheaf. We now turn to more example of sheaves.

Example 3.11 (Sheaves of Continuous and Smooth Functions). Let X and Y be topological spaces. Define $\mathcal{C}$ to be the presheaf of continuous functions, that is, for every open subset $U \subseteq X$,

$$\mathcal{C}(U) := \{f: U \rightarrow Y : f \text{ is continuous}\}.$$

The restriction maps are restrictions of functions.

In a similar manner, let M be a smooth manifold. Then the presheaf of real-valued smooth functions on M (Example 3.2) is a sheaf.

Example 3.12 (Skyscraper Sheaves). Let X be a topological space and A an Abelian group. The **skyscraper sheaf in the point $p \in X$ with value A** is defined to be the sheaf

$$\mathcal{F}(U) = \begin{cases} A, & p \in U; \\ 0, & p \notin U. \end{cases}$$

Not every presheaf is a sheaf, as the following non-examples show.

Example 3.13. Constant presheaves are in general not a sheaf. As an example, consider the two-point topological space $X = \{x, y\}$ with discrete topology. The constant presheaf $\mathcal{F}$ with value $\mathbb{Z}$ is not a sheaf, as the reader may check.

Note that, even if we modify $\mathcal{F}(\emptyset)$ to be equal to $\{0\}$, it is still not a sheaf because the gluing axiom cannot be satisfied.

Example 3.14 (Presheaf of Bounded Functions). Let X be a topological space. For each open set $U \subseteq X$, define

$$\mathcal{F}(U) := \{f: U \rightarrow \mathbb{R} : f \text{ is bounded}\}.$$

Then $\mathcal{F}$ is a presheaf on X . In general it will not be a sheaf.

► **Exercise 3.15.** Verify all the examples.

Sheaves on a Basis

Let X be a topological space and $\mathcal{B}$ a basis of X . It will be useful, when we define affine schemes, to consider sheaves defined explicitly only on the basis $\mathcal{B}$. The main result of this section is that there is a unique sheaf on the whole topological space X extending the sheaf defined on a basis.

Definition 3.16. Suppose X is a topological space and $\mathcal{B}$ a basis of X .

1. A **presheaf on $\mathcal{B}$** is a contravariant functor $\mathcal{F}: \mathcal{B} \rightarrow (\text{Ab})$ where $\mathcal{B}$ is regarded as a full subcategory of (Ouv_X) .
2. A **sheaf on $\mathcal{B}$** is a presheaf $\mathcal{F}$ on $\mathcal{B}$ which satisfies the analogous conditions about gluing as for sheaves: for all $U \in \mathcal{B}$ and an open covering $U = \bigcup_{i \in I} U_i$ where $U_i \in \mathcal{B}$,
 - (a) (Uniqueness) Let $s, t \in \mathcal{F}(U)$ be such that $s|_{U_i} = t|_{U_i}$ for all i . Then $s = t$.
 - (b) (Gluing) A collection of sections $s_i \in \mathcal{F}(U_i)$ is said to be **compatible** if for all $i, j \in I$, and all open set $U_k \in \mathcal{B}$ with $U_k \subseteq U_i \cap U_j$, we have

$$s_i|_{U_k} = s_j|_{U_k}.$$

It is required that, whenever the sections $s_i \in \mathcal{F}(U_i)$ are compatible, then there is a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Note that the gluing condition has to be modified from the gluing condition for sheaves. This is because the intersection of two basic open sets $U_i, U_j \in \mathcal{B}$ might not belong in $\mathcal{B}$. It is still open, however, and so we may cover the intersection $U_i \cap U_j$ by open sets from $\mathcal{B}$. So we need to check compatibility of sections in those smaller open sets instead.

Remark 3.17. If the basis $\mathcal{B}$ is stable under finite intersection (as is the case for basis consisting of principal open subsets), then the usual gluing condition can be used instead. For such a basis $\mathcal{B}$, a presheaf $\mathcal{F}$ on $\mathcal{B}$ is a sheaf if for all $U \in \mathcal{B}$ and an open covering $U = \bigcup_{i \in I} U_i$ where $U_i \in \mathcal{B}$,

1. it satisfy uniqueness, and,
2. for all sections $s_i \in \mathcal{F}(U_i)$, if $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there is a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Proposition 3.18. *Let X be a topological space, $\mathcal{B}$ a basis of X , and $\mathcal{F}$ a sheaf on $\mathcal{B}$. There is a sheaf $\mathcal{F}'$ on X such that $\mathcal{F}'(U) \cong \mathcal{F}(U)$ for all $U \in \mathcal{B}$. The sheaf $\mathcal{F}'$ is unique up to isomorphism.*

Proof. content... □

Morphisms between sheaves on a basis can be extended to the morphisms between the extended sheaves.

3.3 Stalks

Let X be a topological space and let $x \in X$ be a point. The collection of Abelian groups $\mathcal{F}(U)$, where U is an open neighbourhood of x , with restriction maps is a direct system. The direct limit of this system leads to the notion of stalks.

Definition 3.19. Let $\mathcal{F}$ be a presheaf on X . The **stalk** at $x \in X$ of $\mathcal{F}$ is the Abelian group

$$\mathcal{F}_x := \varinjlim_{\substack{U \ni x, \\ U \text{ open}}} \mathcal{F}(U).$$

More explicitly, $\mathcal{F}_x$ is the set of equivalence classes of pairs (U, s) where U is an open neighbourhood of x and $s \in \mathcal{F}(U)$. Two pairs (U, s) and (V, t) are definitionally equivalent if there is an open neighbourhood W of x such that $W \subseteq U \cap V$ and $s|_W = t|_W$. Let us denote the equivalence class of (U, s) by $[U, s]$. The group operation on $\mathcal{F}_x$ is defined by

$$[U, s] + [V, t] = [U \cap V, s|_{U \cap V} + t|_{U \cap V}].$$

► **Exercise 3.20.** Prove that the above relation is indeed an equivalence relation on the set of pairs (U, s) , and that the multiplication is well-defined which makes $\mathcal{F}_x$ into an Abelian group.

By the definition of direct limits, for each open neighbourhood U of x there is a natural homomorphism $\mathcal{F}(U) \rightarrow \mathcal{F}_x$ given by $s \mapsto [U, s]$. If U is understood, we shall write s_x for the equivalent class $[U, s]$. The element s_x is called the **germ** of s in x .

If $\mathcal{F}$ and $\mathcal{G}$ are presheaves on X with a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$, then there is an induced homomorphism of Abelian group between stalks

$$\begin{aligned}\varphi_x: \mathcal{F}_x &\longrightarrow \mathcal{G}_x \\ [U, s] &\longmapsto [U, \varphi_U(s)].\end{aligned}$$

This is the same map from the universal property of direct limits.

► **Exercise 3.21.** Show that φ_x is a group homomorphism.

We thus see that taking stalks gives a functor from the category of presheaves on X to the category of Abelian groups. Further, if the presheaves take values in sets or rings, then stalks will be sets or rings, respectively.

Lemma 3.22. *Let $\mathcal{F}$ be a sheaf on a topological space X and let $s, t \in \mathcal{F}(X)$ be sections such that $s_x = t_x$ for all $x \in X$. Then $s = t$.*

Proof. For each $x \in X$, there is an open neighbourhood U_x such that $s|_{U_x} = t|_{U_x}$. Since the open sets of the form U_x covers X , this shows that $s = t$. $\square$

Injectivity and bijectivity of morphisms in the data of a morphism of sheaves can be checked on stalks, as the following proposition shows.

Proposition 3.23. *Let X be a topological space, $\mathcal{F}$ and $\mathcal{G}$ sheaves on X , and $\varphi, \psi: \mathcal{F} \rightarrow \mathcal{G}$ morphisms of sheaves. Then*

1. *The homomorphisms $\varphi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ are injective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$ are injective for all $x \in X$.*
2. *The homomorphisms $\varphi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ are bijective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$ are bijective for all $x \in X$.*
3. *$\varphi_x = \psi_x$ for all $x \in X$ if and only if $\varphi = \psi$.*

Remark 3.24. The statement fails for surjections. It is in general not true that $\varphi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ are surjective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$ are surjective for all $x \in X$.

Proposition 3.25. *Let X be a topological space, $\mathcal{F}, \mathcal{G}$ sheaves on X , and $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ a morphism of sheaves. Then*

1. *φ is a monomorphism in $(\text{Ab}(X))$ if and only if φ_x is injective for all $x \in X$.*
2. *φ is an epimorphism in $(\text{Ab}(X))$ if and only if φ_x is surjective for all $x \in X$.*
3. *φ is an isomorphism in $(\text{Ab}(X))$ if and only if φ_x is bijective for all $x \in X$.*

Proof. (1) First assume that φ is a monomorphism. In light of Proposition 3.23, it suffices to show that each morphism $\varphi_U: \mathcal{F} \rightarrow \mathcal{G}$ is injective. Let $\mathcal{K}$ be the kernel sheaf of φ , i.e. $\mathcal{K}(U) = \ker \varphi_U$ for all open $U \subseteq X$. This is a subsheaf of $\mathcal{F}$ (Problem 3.1).

Let $i: \mathcal{K} \rightarrow \mathcal{F}$ be the inclusion morphism, i.e. i_U is the inclusion $\mathcal{K}(U) \hookrightarrow \mathcal{F}(U)$ for all open subset $U \subseteq X$. There is also another morphism of sheaves $0: \mathcal{K} \rightarrow \mathcal{F}$ such that $0_U: \mathcal{K}(U) \rightarrow \mathcal{F}(U)$ is the zero homomorphism for all U . Clearly $\varphi \circ i = 0 = \varphi \circ 0$. Since φ is a monomorphism, $i = 0$. Hence for each open $U \subseteq X$, we see that $a = i_U(a) = 0_U(a) = 0$ for all $a \in \ker(\varphi_U)$. In other words, φ_U is injective for all open subset U .

Conversely, assume that φ_x is injective for all $x \in X$. Suppose $\psi, \eta: \mathcal{H} \rightarrow \mathcal{F}$ are morphisms of sheaves such that $\varphi \circ \psi = \varphi \circ \eta$. Since taking stalks is a functor, $\varphi_x \circ \psi_x = \varphi_x \circ \eta_x$ for all $x \in X$. Using the fact that φ_x is injective, we have $\psi_x = \eta_x$ for all $x \in X$. By Proposition 3.23, $\psi = \eta$, and so φ is a monomorphism.

(2)

(3) □

Remark 3.26. Proposition 3.25 remains true if the sheaves considered are sheaves of sets instead of sheaves of Abelian groups, although the arguments must be modified.

Stalks behave well with basis.

Lemma 3.27 (Stalks Can Be Computed on a Basis). *Let X be a topological space with a basis $\mathcal{B}$, $x \in X$ a point, and $\mathcal{F}$ a presheaf on X . The collection of Abelian groups of the form*

$$\mathcal{F}(B), \quad B \in \mathcal{B},$$

together with the restriction maps of $\mathcal{F}$ form a direct system. Furthermore, the direct limit of this system is equal to the stalk of $\mathcal{F}$ at x :

$$\mathcal{F}_x = \varinjlim_{\substack{B \ni x, \\ B \in \mathcal{B}}} \mathcal{F}(B).$$

3.4 Sheafification

As seen above, some presheaves fail to be sheaves because of the failure to glue sections. There is a canonical way to associate a sheaf to a presheaf.

Theorem 3.28 (Sheafification). *Let X be a topological space and $\mathcal{F}$ a presheaf. There exists a sheaf $\mathcal{F}^\dagger$ (called the **sheafification** of $\mathcal{F}$) with a morphism of presheaves $i: \mathcal{F} \rightarrow \mathcal{F}^\dagger$ such that if $\mathcal{G}$ is a sheaf with a morphism of presheaves $\varphi: \mathcal{F} \rightarrow \mathcal{G}$, then there is a unique morphism of sheaves $\varphi^\dagger: \mathcal{F}^\dagger \rightarrow \mathcal{G}$ such that the following diagram commutes:*

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{i} & \mathcal{F}^\dagger \\ & \searrow \varphi & \downarrow \varphi^\dagger \\ & & \mathcal{G} \end{array}$$

The pair $(\mathcal{F}^\dagger, i)$ is unique up to unique isomorphism. Moreover, $i_x: \mathcal{F}_x \rightarrow \mathcal{F}_x^\dagger$ is an isomorphism for all $x \in X$.

The sheaf $\mathcal{F}^\dagger$ is also called the **sheaf associated to the presheaf $\mathcal{F}$** .

Proof. We define the proof into several parts.

[Definition.] Let $U \subseteq X$ be an open subset. Define $\mathcal{F}^\dagger(U)$ to be the set of functions $f: U \rightarrow \coprod_{x \in U} \mathcal{F}_x$ satisfying

- (*) for all $x \in U$, there is an open neighbourhood $V \subseteq U$ of x and $s \in \mathcal{F}(V)$ such that for all $y \in V$, $f(y) = s_y$.

With pointwise addition this is an abelian group. The restriction map $\text{res}_W^U: \mathcal{F}^\dagger(U) \rightarrow \mathcal{F}^\dagger(W)$ for $W \subseteq U$ is defined to be the restriction of functions, so $\text{res}_W^U(f) = f|_W$ for all $f \in \mathcal{F}^\dagger(U)$. It is easily checked that $f|_W \in \mathcal{F}^\dagger(W)$, and that restrictions are group homomorphisms. From this it follows that $\mathcal{F}^\dagger$ is a presheaf on X .

[$\mathcal{F}^\dagger$ is a sheaf.] To show that $\mathcal{F}^\dagger$ is a sheaf, we assume that $U \subseteq X$ is an open set with an open covering $\bigcup_{i \in I} U_i = U$.

1. (Uniqueness) Easy.
2. (Gluing.) For each $i \in I$, let $f_i \in \mathcal{F}^\dagger(U_i)$ be functions such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all $i, j \in I$. Define $f: U \rightarrow \coprod_{x \in U} \mathcal{F}_x$ by $f(x) = f_i(x)$ if U_i contains x . This is well-defined because if U_i and U_j both contains x , then $x \in U_i \cap U_j$ and so $f_i(x) = f_i|_{U_i \cap U_j}(x) = f_j|_{U_i \cap U_j}(x) = f_j(x)$. Further, the condition (*) hold for f .

[Definition of i .] Next we define a morphism of presheaves $i: \mathcal{F} \rightarrow \mathcal{F}^\dagger$. For each $a \in \mathcal{F}(U)$, let $i_U(a)$ be the function $x \mapsto a_x$ for all $x \in U$. This is clearly seen to be a group homomorphism compatible with restriction maps of $\mathcal{F}$ and $\mathcal{F}^\dagger$. Thus i is really a morphism of presheaves.

[i_x is surjective.] Let $x \in X$. We now show that $\mathcal{F}_x^\dagger \cong \mathcal{F}_x$ under the map i_x . Let U be an open neighbourhood of x and let $f \in \mathcal{F}^\dagger(U)$. By definition there is an open neighbourhood V of x and $s \in \mathcal{F}(V)$ such that for all $y \in V$, we have $f(y) = s_y$. Then $[U, f] = [V, i_V(s)]$. In particular, the map $i_x: \mathcal{F}_x \rightarrow \mathcal{F}_x^\dagger$ is a surjection.

[i_x is injective.] To show that i_x is injective, let 0 denote the function $x \mapsto 0_x$ for all $x \in X$. Plainly, $[X, 0]$ is the additive identity of $\mathcal{F}_x$. Suppose that V is a neighbourhood of x and $s \in \mathcal{F}(V)$ is a section such that $i_x([V, s]) = [V, i_V(s)] = [X, 0]$. By definition, there is an open neighbourhood $W \subseteq V$ of x such that $i_W(s|_W) = i_V(s)|_W = 0|_W$. But this implies that $s_y = 0_y$ for all $y \in W$. By Lemma 3.22, $s|_W = 0$. Hence $[V, s] = [W, s|_W] = [W, 0]$, and so i_x is an isomorphism.

[**Universal property of $\mathcal{F}^\dagger$.**] Now we show the universal property. Let $\mathcal{G}$ be a sheaf and $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ a morphism of presheaves. Consider the diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{i} & \mathcal{F}^\dagger \\ \varphi \downarrow & & \downarrow \\ \mathcal{G} & \xrightarrow{j} & \mathcal{G}^\dagger \end{array}$$

where $\mathcal{G}^\dagger$ is defined as set of functions similar to $\mathcal{F}^\dagger$, and j is morphism of presheaves $\mathcal{G} \rightarrow \mathcal{G}^\dagger$. We have shown above that j_x is an isomorphism for all $x \in X$, and so j is an isomorphism of sheaves (Proposition 3.25).

Since we are given a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$, we have the induced morphisms on stalks $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$ for all $x \in X$. Define a morphism $\varphi^\sharp: \mathcal{F}^\dagger \rightarrow \mathcal{G}^\dagger$ on an open set $U \subseteq X$ by

$$\varphi_U^\sharp(f) = \text{the composition } U \xrightarrow{f} \prod_{x \in U} \mathcal{F}_x \xrightarrow{\prod_x \varphi_x} \prod_{x \in U} \mathcal{G}_x,$$

that is, $\varphi_U^\sharp(f)(x) = \varphi_x(f(x))$ for all $x \in U$. It is easy to check that this is a group homomorphism. By composing with j^{-1} we get a morphism $\varphi^\dagger = j^{-1} \circ \varphi^\sharp: \mathcal{F}^\dagger \rightarrow \mathcal{G}$.

To verify that $\varphi^\dagger \circ i = \varphi$, let $U \subseteq X$ be an open set and $s \in \mathcal{F}(U)$. Then $\varphi_U^\sharp(i_U(s))$ is equal to the function given by $s \mapsto \varphi_U(s)_x$ for all $x \in X$. Because j_U is an isomorphism, we immediately have $j_U^{-1}(\varphi_U^\sharp(i_U(s))) = \varphi_U(s)$. This proves the claim.

The fact that $\mathcal{F}^\dagger$ and $\varphi^\dagger$ is unique follows from a standard uniqueness proof of universal objects. $\square$

The most important fact about sheafification can be phrased in the following form: it is a left adjoint to the inclusion functor of sheaves into the category of presheaves.

Proposition 3.29 (Sheafification is left adjoint to the inclusion functor). *Let X be a topological space. Sheafification defines a functor $(-)^{\dagger}$ from the category of presheaves ($\text{PSh}(X)$) to the category of sheaves ($\text{Sh}(X)$). Furthermore, the sheafification functor is left adjoint to the inclusion functor.*

Proof. Follows from Theorem 3.28 $\square$

3.5 Direct Image and Inverse Image

3.6 Locally Ringed Spaces

Definition 3.30. A **ringed space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X .

A **morphism of ringed spaces** $(f, f^\flat): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a pair consisting of a continuous function $f: X \rightarrow Y$ and a morphism of sheaves $f^\flat: \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$.

3.7 Affine Schemes

Let A be a ring. We are going to make $X = \text{Spec } A$ into a locally ringed space by constructing a sheaf of rings on X . Recall that X has a basis consisting of principal open subsets. Define a sheaf $\mathcal{O}_X$ on this basis by setting

$$\mathcal{O}_X(D(f)) = A_f \quad \text{for each } f \in A,$$

where A_f is the localisation of A with respect to the set $\{f^i : i \geq 0\}$. Let us denote the canonical homomorphism $A \rightarrow A_f$ by i_f . To see that this is well-defined, let $D(f) = D(g)$. By Lemma 2.15, $D(g) \subseteq D(f)$ if and only if $g^n = fa$ for some $a \in A$ and $n \geq 1$. In particular, $f/1$ is invertible in A_g because $(a/g^n)(f/1) = 1$. By the universal property of localisation, there exists a unique ring homomorphism $\rho_g^f: A_f \rightarrow A_g$ such that $\rho_g^f \circ i_f = i_g$.

Suppose $D(h) \subseteq D(g) \subseteq D(f)$. Then by uniqueness, $\rho_h^g \circ \rho_g^f = \rho_h^f$. In particular, if $D(g) = D(f)$, then ρ_g^f is an isomorphism which allows us to identify A_f with A_g . The condition $\rho_h^g \circ \rho_g^f = \rho_h^f$ implies that $\mathcal{O}_X$ is a presheaf of rings on $\mathcal{B}$ with the restriction maps given by ρ_g^f .

Theorem 3.31. *The presheaf $\mathcal{O}_X$ is a sheaf on $\mathcal{B}$.*

Proof. Let $D(f)$ be a principal open subset and let $\{D(f_i) : i \in I\}$ be an open cover of $D(f)$. By restricting the presheaf $\mathcal{O}_X$ to $D(f)$, and replacing A_f with A , we may assume that $f = 1$, and hence $D(f) = X$, to ease the notation. Therefore the assumption translates to the statement that $X = \text{Spec } A$ is covered by $\{D(f_i) : i \in I\}$.

We first prove the sheaf conditions when I is finite.

1. Let $s \in \mathcal{O}_X(X) = A$ satisfy $s|_{D(f_i)} = 0$ for all $i \in I$. For each i there is $n_i \in \mathbb{N}$ such that $f_i^{n_i} s = 0$. Set $n = \max_{i \in I} n_i$, so that $f_i^n s = 0$.

Since $D(f_i) = D(f_i^n)$ for all i , it follows that $\bigcup_{i \in I} D(f_i^n) = \bigcup_{i \in I} D(f_i) = X$ and so $\sum_{i \in I} f_i^n A = A$. Thus there are $b_i \in A$ such that $\sum_{i \in I} f_i^n b_i = 1$. This gives

$$s = \left(\sum_{i \in I} f_i^n b_i \right) s = \sum_{i \in I} f_i^n b_i s = 0.$$

2. Let $s_i \in \mathcal{O}_X(D(f_i))$ satisfy $s_i|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)}$ for all $i, j \in I$. Let us write $s_i = \frac{a_i}{f_i^{n_i}}$. We may adjust the fractions so that all n_i are equal, so that $s_i = \frac{a_i}{f_i^n}$, where $n \in \mathbb{N}$ is independent of i .

By hypothesis, for all $i, j \in I$,

$$\frac{a_i}{f_i^n} \Big|_{D(f_i f_j)} = \frac{a_i f_j^n}{(f_i f_j)^n} = \frac{a_j f_i^n}{(f_i f_j)^n} = \frac{a_j}{f_j^n} \Big|_{D(f_i f_j)}.$$

So that there is $m \geq 1$ such that $(f_i f_j)^m (a_i f_j^n - a_j f_i^n) = 0$. Again we may choose m to be independent of i and j . Thus by setting $k = n + m$, we obtain

$$(f_i^m a_i) f_j^k = (f_j^m a_j) f_i^k$$

for all $i, j \in I$.

Arguing as above, there are $b_i \in A$ such that $\sum_{j \in I} f_j^k b_j = 1$. Put $s := \sum_{j \in I} b_j f_j^m a_j$. Then

$$f_i^k s = f_i^k \left(\sum_{j \in I} b_j f_j^m a_j \right) = \sum_{j \in I} b_j (f_i^k f_j^m a_j) = \sum_{j \in I} b_j (f_i^m a_i f_j^k) = f_i^m a_i \left(\sum_{j \in I} b_j f_j^k \right) = f_i^m a_i.$$

Hence $s \in A = \mathcal{O}_X(X)$ is such that its image in A_{f_i} is equal to

$$\frac{s}{1} = \frac{f_i^k s}{f_i^k} = \frac{f_i^m a_i}{f_i^{n+m}} = \frac{a_i}{f_i^n} = s_i.$$

Finally we consider the case when I is infinite. Via Lemma 2.15, there is a finite subset $J \subseteq I$ such that $1 = \sum_{j \in J} f_j A$. From what was just proved, we see that gluing sections indexed by finite subsets work.

1. Let $s \in \mathcal{O}_X(X) = A$ satisfy $s|_{D(f_i)} = 0$ for all $i \in I$. Then $s|_{D(f_j)} = 0$ for all $j \in J$ and hence $s = 0$.
2. Let $s_i \in \mathcal{O}_X(D(f_i))$ satisfy $s_i|_{D(f_i) \cap D(f_k)} = s_k|_{D(f_i) \cap D(f_k)}$ for all $i, k \in I$. Consider the sections s_j with $j \in J$. From the finite case, they glue to a section $s \in \mathcal{O}_X(X)$ such that $s|_{D(f_j)} = s_j$ for all $j \in J$. We claim that the section s is the one we get from gluing s_i .

We only need to check that for all $i \in I$, $s|_{D(f_i)} = s_i$. Note that $\{D(f_i) \cap D(f_j)\}_{j \in J}$ is a finite covering of $D(f_i)$. Hence, for all $i \in I$ and $j \in J$,

$$s|_{D(f_i)}|_{D(f_i) \cap D(f_j)} = s|_{D(f_j)}|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)} = s_i|_{D(f_i) \cap D(f_j)}.$$

By (1), $s|_{D(f_i)} = s_i$. This finishes the proof. $\square$

Corollary 3.32. *The sheaf $\mathcal{O}_X$ extends to a sheaf $\mathcal{O}_X$ on X , called the **structure sheaf** of $\text{Spec } A$.*

Lemma 3.33 (Stalks of $\text{Spec } A$). *Let $X = \text{Spec } A$, $\mathcal{O}_X$ its structure sheaf, and $x \in \text{Spec } A$. Then*

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}_x},$$

the localisation of A at the prime ideal $\mathfrak{p}_x$. In particular, $(X, \mathcal{O}_X)$ is a locally ringed space.

Proof. By Lemma 3.27, we can compute the stalk $\mathcal{O}_{X,x}$ on the basis of principal open subsets, and thus

$$\mathcal{O}_{X,x} = \varinjlim_{D(f) \ni x} \mathcal{O}_X(D(f)) = \varinjlim_{f \notin \mathfrak{p}} A_f = A_{\mathfrak{p}},$$

the last equality follows from commutative algebra (Problem 3.2). $\square$

Definition 3.34. A locally ringed space $(X, \mathcal{O}_X)$ is said to be an **affine scheme** if it is isomorphic to a locally ringed space $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for suitable ring A .

A **morphism of affine schemes** is simply a morphism of locally ringed spaces. We thus obtain the category of affine schemes which we shall denote by (Aff) .

Let $\varphi: A \rightarrow B$ be a ring homomorphism. We shall define a morphism of affine schemes $\text{Spec } \varphi = (f, f^\flat): (\text{Spec } B, \mathcal{O}_{\text{Spec } B}) \rightarrow (\text{Spec } A, \mathcal{O}_{\text{Spec } A})$. Here is the definition:

1. $f = \text{Spec } \varphi$ is the induced continuous map on spectra, and,
2. $f^\flat: \mathcal{O}_{\text{Spec } A} \rightarrow f_* \mathcal{O}_{\text{Spec } B}$ is given on principal open subsets by

$$f_{D(s)}^\flat: \mathcal{O}_{\text{Spec } A}(D(s)) \rightarrow \mathcal{O}_{\text{Spec } B}(f^{-1}(D(s))).$$

Since $f^{-1}(D(s)) = D(\varphi(s))$ by Problem 2.7, we need to define a map $f_{D(s)}^\flat: A_s \rightarrow B_{\varphi(s)}$. The natural map for this is the unique map $\varphi_s: A_s \rightarrow B_{\varphi(s)}$ induced by the canonical localisation maps:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ A_s & \xrightarrow{\varphi_s} & B_{\varphi(s)} \end{array}$$

Therefore we take $f_{D(s)}^\flat$ to be φ_s .

The above data actually define a morphism of ringed spaces $\text{Spec } B \rightarrow \text{Spec } A$, as shown in the following proposition.

Proposition 3.35. *The maps $f_{D(s)}^\flat$ is compatible with restrictions, and hence define a morphism of sheaves.*

Anti-equivalence of the Categories (Aff) and (Ring)

3.8 Problems

- 3.1** Prove that if $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves on a topological space X , then the **kernel sheaf** $\mathcal{K}$ defined on each open subset $U \subseteq X$ by the formula

$$\mathcal{K}(U) = \ker(\varphi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

is a subsheaf of $\mathcal{F}$. (The restriction maps of $\mathcal{K}$ is inherited from those of $\mathcal{F}$.)

- 3.2** Let A be a ring and $\mathfrak{p}$ a prime ideal. Show that the rings $(A_f)_{f \notin \mathfrak{p}}$ together with homomorphisms $A_f \rightarrow A_g$ whenever $g \in \sqrt{fA}$ form a direct system. Show that the limit of this direct system is $A_{\mathfrak{p}}$

Chapter 4

Schemes

4.1 Schemes

After defining affine schemes, we can now define schemes as locally ringed spaces that are affine schemes glued together.

Definition 4.1. A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ which admits a covering by open subsets $\{U_i\}_{i \in I}$ such that $(U_i, \mathcal{O}_{X|U_i})$ is isomorphic to an affine scheme for all $i \in I$.

A **morphism of schemes** is a morphism of the underlying locally ringed spaces. Schemes and morphisms of schemes together form the category of schemes, which is denoted by (Sch) . The category of affine schemes is then a full subcategory of (Sch) . We shall often write X for the scheme $(X, \mathcal{O}_X)$; its structure sheaf is left implicit.

Schemes Over a Base Scheme

Generalising the notion of varieties defined over a field k , there is a corresponding notion of schemes over a base scheme. Let S be a scheme. An **S -scheme** is a scheme X with a morphism of schemes $X \rightarrow S$, called the **structural morphism** of X . Given two S -schemes $X \rightarrow S$ and $Y \rightarrow S$, a **morphism of S -schemes** $f: X \rightarrow Y$ is a morphism of schemes such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow & \swarrow \\ & S & \end{array}$$

commutes. The scheme S is called the **base scheme**. It is itself an S -scheme with the morphism $\text{id}_S: S \rightarrow S$. In this manner, we have the **category of S -schemes** (Sch/S) . Another way of viewing (Sch/S) is that it is the slice category of (Sch) over S . The set of morphisms between X and Y in the category of S -schemes will be denoted by $\text{Hom}_S(X, Y)$, to avoid confusion with the hom-set $\text{Hom}(X, Y)$ when X, Y are just regarded as schemes.

In many cases when the base scheme is affine, we shall employ a common shorthand notation. If S is affine, say $S = \operatorname{Spec} R$, a scheme over S is also called an **R -scheme** or a **scheme over R** . The hom-set of R -schemes X and Y is written by $\operatorname{Hom}_R(X, Y)$.

Open Subschemes

4.2 Gluing Lemmas

Lemma 4.2 (Morphism Gluing Lemma). *Let X and Y be locally ringed spaces, and let $\{U_i\}_{i \in I}$ be an open covering of X . Then a family of morphisms $f_i: U_i \rightarrow Y$ of locally ringed spaces together form a morphism $f: X \rightarrow Y$ if and only if the morphisms $f_i: U_i \rightarrow Y$ and $f_j: U_j \rightarrow Y$ coincide on $U_i \cap U_j$ for all $i, j \in I$. Furthermore, the resulting morphism $X \rightarrow Y$ is unique.*

Proof. The only non-trivial part is constructing a morphism $f: X \rightarrow Y$ from the morphisms $f_i: U_i \rightarrow Y$. At the level of the underlying topological spaces this is always possible, so we obtain a continuous map $f: X \rightarrow Y$ such that $f|_{U_i} = f_i$ for all $i \in I$.

We seek to define a morphism of sheaves $f^\flat: \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. Let $V \subseteq Y$ be an open subset, so that $f^{-1}(V)$ is an open subset of X . For each $i \in I$, we are given a map

$$\mathcal{O}_Y(V) \rightarrow (f_i)_*\mathcal{O}_{X|U_i}(V) = \mathcal{O}_X(U_i \cap f^{-1}(V)).$$

For each $s \in \mathcal{O}_Y(V)$, let s_i denote the image of s in $\mathcal{O}_X(U_i \cap f^{-1}(V))$ under the above map. The idea is to glue the sections s_i together and take the resulting section to be $f_V^\flat(s)$. To make sure that the sections can be glued together, we need to show that

$$s_i|_{U_i \cap U_j \cap f^{-1}(V)} = s_j|_{U_i \cap U_j \cap f^{-1}(V)}$$

for all $i, j \in I$. This is implied by our assumption that f_i and f_j coincide on $U_i \cap U_j$. Furthermore, $f_V^\flat(s)$ is well-defined by the uniqueness condition of sheaf axioms.

The only thing left to check is that the maps $f_V^\flat$ are compatible with restriction maps. Let $W \subseteq V$ be open subsets of Y . Then for all $s \in \mathcal{O}_Y(V)$ and $i \in I$,

$$\begin{aligned} f_V^\flat(s)|_{f^{-1}(W) \cap U_i} &= f_V^\flat(s)|_{f^{-1}(V) \cap U_i}|_{f^{-1}(W) \cap U_i} \\ &= s_i|_{f^{-1}(W) \cap U_i} \\ &= (s|_W)_i. \end{aligned}$$

By uniqueness, $f_V^\flat(s)|_{f^{-1}(W)} = f_W^\flat(s|_W)$. This proves that $f^\flat$ is compatible with restriction maps and so define a morphism of sheaves $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. For each $x \in X$, the induced map on stalks $f_x^\sharp$ is plainly a local ring homomorphism because taking stalk is local. $\square$

Lemma 4.3 (Scheme Gluing Lemma). *content...*

4.3 Immersions

4.4 Problems

Chapter 5

Properties of Schemes

5.1 Noetherian Schemes

Recall that a ring R is **Noetherian** if it satisfies the *ascending ideal condition*: every chain of ideals

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

is eventually constant, that is, there is $N \in \mathbb{N}$ such that $I_N = I_n$ for all $n \geq N$. Equivalently every ideal of R is finitely generated. The corresponding notion for schemes is as follows.

Definition 5.1. A scheme X is **locally Noetherian** if X admits an open covering $X = \bigcup_i U_i$ by affine open subschemes U_i such that the ring $\mathcal{O}_X(U_i)$ is Noetherian. A scheme is **Noetherian** if it is quasi-compact and locally Noetherian.

Recall that every affine scheme is quasi-compact. Hence locally Noetherian affine scheme is automatically Noetherian. Since localisations of a Noetherian ring are again Noetherian, this means that every locally Noetherian scheme has a basis consisting of Noetherian affine open subschemes.

Proposition 5.2. *The following are equivalent for a scheme X .*

1. X is locally Noetherian.
2. Each point $x \in X$ has an affine open neighbourhood U such that $\mathcal{O}_X(U)$ is Noetherian.
3. For every affine open subset $U \subseteq X$ the ring $\mathcal{O}_X(U)$ is Noetherian.

Proof. Clearly (1) $\iff$ (2) and (3) $\Rightarrow$ (1) because schemes are covered by affine open subsets. In turn we only need to show that (1) implies (3).

Let $U = \text{Spec } R$ be an affine open subscheme. By the observation previously discussed, we may assume that U can be covered by Noetherian affine open subschemes U_i . $\square$

Chapter 6

Schemes and Varieties

Chapter 7

Projective Spaces

Chapter 8

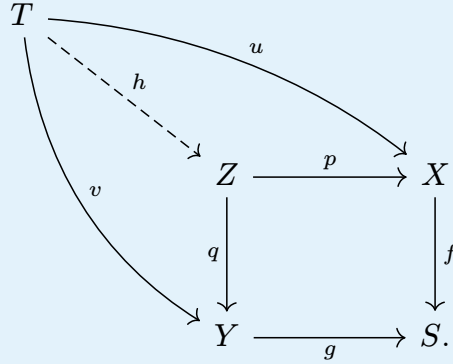
Fibre Products

8.1 Fibre Products

We start with the definition of fibre products in any category.

Definition 8.1. Let $\mathcal{C}$ be a category, and let $f: X \rightarrow S$ and $g: Y \rightarrow S$ be morphisms in $\mathcal{C}$. A **fibre product** or **pullback** of f and g is an object Z with morphisms $p: Z \rightarrow X$ and $q: Z \rightarrow Y$ which is universal with respect to the following property:

The equation $f \circ p = g \circ q$ holds; and furthermore, for any object T with morphisms $u: T \rightarrow X$ and $v: T \rightarrow Y$ such that $f \circ u = g \circ v$, there exists a unique morphism $h: T \rightarrow Z$ such that the following diagram commutes



A fibre product of f and g is also called a **fibre product of X and Y over S** . As with any universal objects, the fibre product of f and g is unique up to unique isomorphism if it exists. Thus we may denote the fibre product Z of f and g by $X \times_S Y$. The morphisms $X \times_S Y \rightarrow X$ and $X \times_S Y \rightarrow Y$ are called the **projections** to the first and second factor, respectively.

The unique morphism h arising from u and v as in the definition of fibre product will often be denoted by $(u, v)_S$. The notation emphasises the fact that $p \circ (u, v)_S = u$ and $q \circ (u, v)_S = q$, showing that p and q acts like projections.

For familiar categories such as the category of sets and the category of topological

spaces, fibre products always exist and can be explicitly described.

Example 8.2. Let $f: X \rightarrow S$ and $g: Y \rightarrow S$ be functions. Then the set

$$Z = \{(x, y) \in X \times Y : f(x) = g(y)\},$$

together with projection maps $p: (x, y) \mapsto x$ and $q: (x, y) \mapsto y$, is the fibre product of f and g in (Set).

For the category of topological spaces, equip the set Z above with the subspace topology induced from the product topology of $X \times Y$. The projection maps p and q are then continuous, and together with Z form the fibre product of f and g in (Top).

Keeping the notation of Example 8.2, the fibre product of f and g can also be written as

$$Z = X \times_S Y = \bigcup_{s \in S} f^{-1}(s) \times g^{-1}(s),$$

hence the name *fibre product*.

► **Exercise 8.3.** For any category $\mathcal{C}$ and object $S \in \mathcal{C}$, prove that the fibre product of $f: X \rightarrow S$ and $Y \rightarrow S$ is the same as the product of f and g taken in the slice category $\mathcal{C}/S$.

Functoriality

We now turn to functoriality of fibre products. Suppose that in the category $\mathcal{C}$ all fibre products exist. Given morphisms $f: X \rightarrow S$ and $g: Y \rightarrow S$ we may form the fibre product $X \times_S Y$:

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p} & X \\ \downarrow q & & \downarrow f \\ Y & \xrightarrow{g} & S. \end{array}$$

Now suppose X' and Y' are S -objects with S -morphisms $u: X' \rightarrow X$ and $v: Y' \rightarrow Y$. Again we make form the fibre product $(X' \times_S Y', p', q')$. Then there is a unique morphism

$u \times_S v: X' \times_S Y' \rightarrow X \times_S Y$ such that the diagram

$$\begin{array}{ccccc}
 X' \times_S Y' & \xrightarrow{p'} & X' & & \\
 \downarrow q' & \searrow u \times_S v & \searrow u & & \\
 & & X \times_S Y & \xrightarrow{p} & X \\
 & \searrow v & \downarrow q & & \downarrow f \\
 & & Y & \xrightarrow{g} & S
 \end{array}$$

is commutative. In fact, $u \times_S v = (u \circ p', v \circ q')_S$ is obtained by the universal property of fibre products, as one can easily check.

The following proposition is easy to prove either directly or via the Yoneda lemma.

Proposition 8.4. *Let S be an object in $\mathcal{C}$ and let X, Y, Z be S -objects. Then there are “canonical” isomorphisms*

1. $X \times_S S \xrightarrow{\sim} X$.
2. $X \times_S Y \xrightarrow{\sim} Y \times_S X$.
3. $X \times_S (Y \times_S Z) \xrightarrow{\sim} (X \times_S Y) \times_S Z$.

All of the isomorphisms are functorial in X, Y , and Z .

Given a commutative diagram

$$\begin{array}{ccc}
 Z & \xrightarrow{p} & X \\
 \downarrow q & & \downarrow f \\
 Y & \xrightarrow{g} & S
 \end{array}$$

we say that this diagram is **cartesian** if Z and the morphisms p, q is the fibre product of f and g . We denote a cartesian diagram using a square $\square$ in the middle as shown below:

$$\begin{array}{ccc}
 Z & \xrightarrow{p} & X \\
 \downarrow q & \square & \downarrow f \\
 Y & \xrightarrow{g} & S
 \end{array}$$

We finish this section with the *pullback lemma*.

Lemma 8.5 (Pullback lemma). *Suppose we are given the following commutative diagram in a category $\mathcal{C}$*

$$\begin{array}{ccccc}
 Z' & \xrightarrow{p'} & Z & \xrightarrow{p} & X \\
 \downarrow q' & & \downarrow q & \square & \downarrow f \\
 Y' & \xrightarrow{g'} & Y & \xrightarrow{g} & S
 \end{array}$$

in which the right square is cartesian. Then the left square is cartesian if and only if the outer rectangle is cartesian.

► **Exercise 8.6.** Prove the pullback lemma either by diagram chasing or by the Yoneda lemma.

8.2 Fibre Product of Schemes

The goal of this section is to prove that fibre products always exists in the category of schemes. We begin by showing that if all the schemes X , Y , and S involved are affine, then the fibre product of X and Y over S exists.

Theorem 8.7. *Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$ and $S = \operatorname{Spec} R$ be affine schemes, and let $f: X \rightarrow S$ and $g: Y \rightarrow S$ be morphisms of schemes. Then the fibre product of f and g exists and is given by the affine scheme $Z = \operatorname{Spec}(A \otimes_R B)$. The projections $p: Z \rightarrow X$ and $q: Z \rightarrow Y$ are the morphisms corresponding to the ring homomorphisms*

$$\begin{aligned}
 \alpha: A &\rightarrow A \otimes_R B, & a &\mapsto a \otimes 1, \\
 \beta: B &\rightarrow A \otimes_R B, & b &\mapsto 1 \otimes b.
 \end{aligned}$$

Proof. Recall that a morphism of affine schemes $f: X \rightarrow S$ corresponds to a ring homomorphism $\varphi: R \rightarrow A$. Similarly, g corresponds to a ring homomorphism $\psi: R \rightarrow B$. Hence A and B are R -algebras and so the tensor product $A \otimes_R B$ exists. $\square$

Theorem 8.8 (Fibre Product of Schemes). *Let S be a scheme and let X, Y be S -schemes. Then the fibre product $X \times_S Y$ exists in the category of schemes.*

Chapter 9

Schemes over Fields

9.1 Dimension Theory

Chapter 10

Separated Morphisms

Recall that a topological space X is Hausdorff if for every two distinct points $x \neq y$, there are open neighbourhoods U of x and V of y such that $U \cap V = \emptyset$. The following conditions are equivalent to X being Hausdorff:

1. The diagonal set $\Delta := \{(x, x) : x \in X\}$ is closed in $X \times X$, where $X \times X$ is equipped with the product topology.
2. For every topological space Y and continuous map $f : Y \rightarrow X$, the graph $\{(y, f(y)) : y \in Y\}$ is closed in $Y \times X$.
3. For every topological space Y and continuous maps $f, g : Y \rightarrow X$, the equaliser $\{y \in Y : f(y) = g(y)\}$ is closed in Y .

► **Exercise 10.1.** Prove the asserted equivalence.

Schemes are in general not Hausdorff. Nevertheless, the above characterisation serves as a motivation for an analogue of Hausdorff property of schemes.

10.1 Separated Morphisms

Diagonals, Graphs, and Equalisers

We first need to define diagonals, graphs and equalisers for arbitrary schemes. In more generality we shall define them for any category which has fibre products.

In this section, let $\mathcal{C}$ be a category admitting all fibre products. Let $S \in \mathcal{C}$ and $u : X \rightarrow S$ an S -object. If $T \rightarrow S$ is another S -object, the set of all S -morphisms $T \rightarrow X$ is denoted by $X_S(T)$.

Definition 10.2. Situation as above.

1. The **diagonal** of $u: X \rightarrow S$ is the morphism

$$\Delta_{X/S} := \Delta_u := (\text{id}_X, \text{id}_X)_S: X \rightarrow X \times_S X$$

obtained from the universal property of fibre products. Sometimes we write Δ for $\Delta_{X/S}$ if no confusion can arise

2. Let $f: X \rightarrow Y$ be an S -morphism. The **graph** of f is the morphism

$$\Gamma_f := (\text{id}_X, f)_S: X \rightarrow X \times_S Y.$$

3. Let $f, g: X \rightarrow Y$ be two S -morphisms. The **equaliser** $\text{Eq}(f, g)$ of f, g is the categorical equaliser of f and g in the slice category over S . The universal S -morphism $i: \text{Eq}(f, g) \rightarrow X$ is called the **canonical** morphism.

Unravelling the definition of equalisers, we can give another definition using universal mapping property as follows. The canonical morphism $i: \text{Eq}(f, g) \rightarrow X$ satisfies $f \circ i = g \circ i$ and for all S -morphism $x: T \rightarrow X$ such that $f \circ x = g \circ x$, there is a unique morphism $p: T \rightarrow \text{Eq}(f, g)$ such that $i \circ p = x$. This is indicated in the following commutative diagram:

$$\begin{array}{ccc} \text{Eq}(f, g) & \xrightarrow{i} & X \\ \uparrow p & \nearrow x & \\ T & & \end{array} \quad \begin{array}{c} f \\ \rightrightarrows \\ g \end{array} Y.$$

Alternatively, $i: K \rightarrow X$ is an equaliser of f, g if and only if it represents the contravariant functor sending an S -object T to the set $\{x \in X_S(T) : f(T)(x) = g(T)(x)\}$.

► **Exercise 10.3.** Show that the canonical morphism $i: \text{Eq}(f, g) \rightarrow X$ is a monomorphism in the category of S -objects.

The following lemma shows that the equaliser of two S -morphisms can be expressed in terms of fibre products, and so all equalisers exist in $\mathcal{C}$.

Lemma 10.4.

Example 10.5. In the category of sets, the diagonal

Diagonals, Graphs, and Equalisers in the Category of Schemes

Proposition 10.6. *Let $S = \operatorname{Spec} R$, $X = \operatorname{Spec} A$ and $\operatorname{Spec} B$.*

Chapter 11

Proper Morphisms

Definition 11.1 (Closed and Universally Closed Morphisms). Let $f: X \rightarrow Y$ be a morphism of schemes.

1. We say that f is **closed** if for every closed subset $Z \subseteq X$, its image $f(Z)$ is closed in Y . In other words, f is closed as a topological map.
2. The morphism f is **universally closed** if for every morphism $Y' \rightarrow Y$, the base change $X \times_Y Y' \rightarrow Y'$ is closed.

Definition 11.2. A morphism $f: X \rightarrow Y$ is **proper** if it is separated, of finite type, and universally closed.

Lemma 11.3. *Let $\mathbf{P}$ be a property of morphisms such that*

1. *$\mathbf{P}$ is stable under composition;*
2. *$\mathbf{P}$ is stable under base change;*
3. *Every closed immersion has $\mathbf{P}$.*

Proposition 11.4. *Closed immersions are proper.*

Chapter 12

Sheaves of Modules

12.1 Sheaves of Modules

Definition 12.1. Let $(X, \mathcal{O}_X)$ be a ringed space. A **sheaf of modules** on X or an **$\mathcal{O}_X$ -module** is an Abelian sheaf $\mathcal{F}$ on X satisfying the following properties:

1. For each open subset U of X , $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module.
2. The module structures are compatible with restriction maps: if $V \subseteq U$ are open subsets, $a \in \mathcal{O}_X(U)$, and $s \in \mathcal{F}(U)$, then

$$(as)|_V = a|_V s|_V.$$

Example 12.2. The sheaf $\mathcal{O}_X$ is trivially an $\mathcal{O}_X$ -module. More generally, if $i \in \mathbb{N}$, then the sheaf

$$U \mapsto \underbrace{\mathcal{O}_X(U) \oplus \cdots \oplus \mathcal{O}_X(U)}_{i \text{ summands}}$$

is an $\mathcal{O}_X$ -module. We will generalise this construction later.

Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}_X$ -modules. A **morphism of $\mathcal{O}_X$ -modules** $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves such that for each open subset $U \subseteq X$, the map on

$$\varphi_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is also an $\mathcal{O}_X(U)$ -module homomorphism. The identity morphism $\text{id}: \mathcal{F} \rightarrow \mathcal{F}$ is immediately seen to be a morphism of $\mathcal{O}_X$ -modules. Moreover, the composition of two morphisms of $\mathcal{O}_X$ -modules is again a morphism of $\mathcal{O}_X$ -modules. We thus obtain the category $(\mathcal{O}_X\text{-Mod})$ of $\mathcal{O}_X$ -modules. The set of all morphisms of $\mathcal{O}_X$ -modules from $\mathcal{F}$ to $\mathcal{G}$ is denoted by $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$.

► **Exercise 12.3.** Show that $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is naturally an $\mathcal{O}_X(X)$ -module. In particular, the hom-set $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is an Abelian group.

► **Exercise 12.4.** Show that, for each $\mathcal{O}_X$ -module $\mathcal{F}$, the stalk $\mathcal{F}_x$ at $x \in X$ has the induced $\mathcal{O}_{X,x}$ -module structure. The multiplication of $[U, a]$ and $[V, s]$ where U, V are open neighbourhood of x , $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{F}(V)$, is defined to be

$$[U, a][V, s] = [U \cap V, a|_{U \cap V} s|_{U \cap V}].$$

Show that if $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of $\mathcal{O}_X$ -modules, then the map of stalks $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$ is a homomorphism of $\mathcal{O}_{X,x}$ -modules.

The result of Exercise 12.4 implies that taking stalks defines a functor from $(\mathcal{O}_X\text{-Mod})$ to $(\mathcal{O}_{X,x}\text{-Mod})$.

Definition 12.5. Let $(X, \mathcal{O}_X)$ be a locally ringed space, $\mathcal{F}$ an $\mathcal{O}_X$ -module, and $x \in X$. By Exercise 12.4, the stalk $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$ -module. The tensor product

$$\mathcal{F}(x) := \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

where $\kappa(x)$ is the residue field of x , is an $\kappa(x)$ -vector space. This is called the **stalk of $\mathcal{F}$ at x** .

We now turn to more examples of $\mathcal{O}_X$ -modules. For the first example, recall that a *zero object* in a category $\mathcal{C}$ is an object which is both initial and final.

Example 12.6 (The Zero Sheaf). The sheaf 0 defined by $0(U) = \{0\}$ for all open subset $U \subseteq X$ is an $\mathcal{O}_X$ -module. For every $\mathcal{O}_X$ -module $\mathcal{F}$, there are unique morphisms $\mathcal{F} \rightarrow 0$ and $0 \rightarrow \mathcal{F}$ making 0 the zero object in $(\mathcal{O}_X\text{-Mod})$.

Example 12.7 (Sheaves of Abelian Groups as Sheaves of Modules). content...

12.2 Constructions on $\mathcal{O}_X$ -modules

12.3 Category of $\mathcal{O}_X$ -modules is Abelian

12.4 Sheaves of Modules Attached to Modules

In this section we show that modules over a ring give rise to sheaves of modules over the spectrum of that ring. Let A be a ring and M an A -module. Set $X = \text{Spec } A$. Recall that the collection

$$\mathcal{B} = \{D(f) : f \in A\}$$

of principal open subsets of X is a basis of X . Define the sheaf $\tilde{M}$ on each principal open set $D(f) \subseteq X$ ($f \in A$) by the formula

$$\tilde{M}(D(f)) = M_f,$$

where M_f is the localisation of M with respect to the multiplicative set $\{f^i : i \geq 0\}$. If $D(f) \subseteq D(g)$, then the restriction map $\tilde{M}(D(g)) \rightarrow \tilde{M}(D(f))$ is taken to be the homomorphism of A_g -modules $M_g \rightarrow M_f$ obtained from the universal property of localisation. The same argument as in the definition of affine schemes show that $\tilde{M}$ is indeed a sheaf on the basis $\mathcal{B}$.

Proposition 12.8. *The assignment $\tilde{M}$ is a sheaf on the basis $\mathcal{B}$.*

Hence $\tilde{M}$ extends to a sheaf on X which we again denote by $\tilde{M}$. As is readily seen from the construction, $\tilde{M}(D(f)) = M_f$ is a module over $\mathcal{O}_X(D(f)) = A_f$. In fact, the module structure extends to the whole of $\tilde{M}$, rendering $\tilde{M}$ into an $\mathcal{O}_X$ -module.

Lemma 12.9. *Let $U \subseteq \operatorname{Spec} A$ be an open subset. Then $\tilde{M}(U)$ carries an $\mathcal{O}_X(U)$ -module structure making $\tilde{M}$ into an $\mathcal{O}_X$ -module.*

Proof. Let $a \in \mathcal{O}_X(U)$ and $s \in \tilde{M}(U)$. We define the multiplication as locally on each principal open subset of U and then show that these local sections can be glued together. Let $D(f) \subseteq U$, $f \in A$ be a principal open subset. Then $a|_{D(f)} \in \mathcal{O}_X(D(f)) = A_f$ and $s|_{D(f)} \in \tilde{M}(D(f)) = M_f$. In particular, the multiplication $a|_{D(f)} s|_{D(f)}$ is defined plainly as the scalar multiplication of M_f .

In order to glue these sections together, we have to check that for two principal open subsets $D(f)$ and $D(g)$ of U ,

$$(a|_{D(f)} s|_{D(f)})|_{D(f) \cap D(g)} = (a|_{D(g)} s|_{D(g)})|_{D(f) \cap D(g)}.$$

Recall that $D(f) \cap D(g) = D(fg)$, and so the restriction map $\tilde{M}(D(f)) \rightarrow \tilde{M}(D(fg))$ is simply the A_f -module homomorphism $M_f \rightarrow M_{fg}$. Similarly the restriction map $\tilde{M}(D(g)) \rightarrow \tilde{M}(D(fg))$ is the A_g -module homomorphism $M_g \rightarrow M_{fg}$. Using the fact that the restriction maps are module homomorphisms, we get

$$\begin{aligned} (a|_{D(f)} s|_{D(f)})|_{D(f) \cap D(g)} &= (a|_{D(f)})|_{D(f) \cap D(g)} (s|_{D(f)})|_{D(f) \cap D(g)} \\ &= a|_{D(fg)} s|_{D(fg)} \\ &= (a|_{D(g)})|_{D(f) \cap D(g)} (s|_{D(g)})|_{D(f) \cap D(g)} \\ &= (a|_{D(g)} s|_{D(g)})|_{D(f) \cap D(g)}. \end{aligned}$$

Hence the sections $a|_{D(f)} s|_{D(f)}$, where $D(f)$ runs over all principal open subsets of U , glue to a unique section $as \in \tilde{M}(U)$. The module axioms for $\tilde{M}(U)$ and the fact that the module structures are compatible with restriction maps can be checked locally in the similar manner as the above. $\square$

By construction, $\tilde{A} = \mathcal{O}_X$.

12.5 Quasi-coherent Modules

Chapter 13

Line Bundles and Divisors

13.1 Cartier Divisors

Chapter 14

Smoothness and Differentials

Chapter 15

Projective Schemes

Chapter 16

Čech Cohomology

Let X be a ringed space and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open covering of X .

Chapter 17

Cohomology of Sheaves of Modules

Chapter 18

Serre Duality

Chapter 19

Cohomology and Base Change

Chapter 20

Grassmannians, Flag Varieties, Schubert Varieties

Chapter 21

Torsors

21.1 Torsors

Let X be a topological space and G a sheaf of groups on X . If T is a sheaf of sets on X , then we say that G **acts on** T or that T is a **G -sheaf** if there is a morphism of sheaves $G \times T \rightarrow T$ such that for every open subset $U \subseteq X$, the map $G(U) \times T(U) \rightarrow T(U)$ is a left group action of $G(U)$ on $T(U)$.

A **morphism of G -sheaves** is a morphism of sheaves $\varphi: T \rightarrow T'$ such that $\varphi_U: T(U) \rightarrow T'(U)$ is $G(U)$ -equivariant for all open subset $U \subseteq X$. By definition, this means that $\varphi_U(g \cdot t) = g \cdot \varphi_U(t)$ for all open subset $U \subseteq X$, $g \in G(U)$, and $t \in T(U)$.

Definition 21.1. A G -sheaf T is called a **G -torsor** if it satisfies the following two properties:

1. The group $G(U)$ acts simply transitively on $T(U)$ for every open subset $U \subseteq X$. (A group G acts *simply transitively* on X if for all $t, s \in X$, there is a unique $g \in G$ such that $g \cdot t = s$.)
2. There exists an open covering $\{U_i\}_{i \in I}$ of X such that $T(U_i) \neq \emptyset$ for all $i \in I$.

An example of a G -torsor is G itself under left multiplication action. It is called the **trivial G -torsor**.

Remark 21.2. A G -torsor T is isomorphic to the trivial G -torsor if and only if $T(X) \neq \emptyset$. To see this, take any $t \in T(X)$ and define a morphism $f_U: G(U) \rightarrow T(U)$ by $g \mapsto gt|_U$ for all open subset $U \subseteq X$ and $g \in G(U)$. The fact that $G(U)$ acts simply transitively implies that each f_U is an isomorphism. The morphisms f_U are compatible with restriction maps and hence together form an isomorphism $f: G \rightarrow T$ of sheaves.

Chapter 22

Curves

Chapter 23

Category Theory